

**Tabela complementar S4.** Métricas de desempenho individuais obtidas na etapa de validação do GroundEye S800, para as amostras de sementes com presença de defeito tipo popped kernel nas proporções de 0%, 2,5%, 7,5%, 15% e 25%

| Método | Perc | Preceça de dano % | Acurácia | Especificidade | Taxa de erro | Precisão | MCC    |
|--------|------|-------------------|----------|----------------|--------------|----------|--------|
| ANOVA  | 0.05 | 0                 | 0.955    | 0.667          | 0.045        | 0.998    | 0.208  |
|        |      | 2.5               | 0.948    | 0.824          | 0.052        | 0.995    | 0.504  |
|        |      | 7.5               | 0.937    | 0.886          | 0.063        | 0.991    | 0.663  |
|        |      | 15                | 0.915    | 0.827          | 0.085        | 0.972    | 0.682  |
|        |      | 25                | 0.945    | 0.947          | 0.055        | 0.982    | 0.861  |
|        | 0.1  | 0                 | 0.975    | 0.667          | 0.025        | 0.998    | 0.282  |
|        |      | 2.5               | 0.967    | 0.824          | 0.033        | 0.995    | 0.595  |
|        |      | 7.5               | 0.96     | 0.886          | 0.04         | 0.991    | 0.752  |
|        |      | 15                | 0.933    | 0.827          | 0.067        | 0.972    | 0.733  |
|        |      | 25                | 0.978    | 0.93           | 0.035        | 0.978    | 0.907  |
|        | 0.2  | 0                 | 0.978    | 0.333          | 0.022        | 0.997    | 0.159  |
|        |      | 2.5               | 0.975    | 0.824          | 0.025        | 0.975    | 0.654  |
|        |      | 7.5               | 0.97     | 0.818          | 0.033        | 0.986    | 0.784  |
|        |      | 15                | 0.958    | 0.889          | 0.042        | 0.982    | 0.829  |
|        |      | 25                | 0.965    | 0.927          | 0.035        | 0.976    | 0.906  |
|        | 0.3  | 0                 | 0.99     | 0.667          | 0.01         | 0.998    | 0.432  |
|        |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.733  |
|        |      | 7.5               | 0.977    | 0.795          | 0.023        | 0.984    | 0.822  |
|        |      | 15                | 0.962    | 0.889          | 0.038        | 0.982    | 0.841  |
|        |      | 25                | 0.973    | 0.92           | 0.027        | 0.974    | 0.9282 |
|        | 0.5  | 0                 | 0.992    | 0.667          | 0.008        | 0.998    | 0.468  |
|        |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.733  |
|        |      | 7.5               | 0.978    | 0.818          | 0.022        | 0.986    | 0.836  |
|        |      | 15                | 0.958    | 0.864          | 0.042        | 0.979    | 0.825  |
|        |      | 25                | 0.977    | 0.933          | 0.023        | 0.978    | 0.9372 |
|        | 0.75 | 0                 | 0.988    | 0.667          | 0.012        | 0.998    | 0.404  |
|        |      | 2.5               | 0.987    | 0.824          | 0.013        | 0.995    | 0.772  |
|        |      | 7.5               | 0.927    | 0.795          | 0.073        | 0.983    | 0.595  |
|        |      | 15                | 0.962    | 0.877          | 0.038        | 0.981    | 0.839  |
|        |      | 25                | 0.973    | 0.92           | 0.027        | 0.974    | 0.928  |
|        | 1    | 0                 | 0.985    | 0.667          | 0.015        | 0.998    | 0.36   |
|        |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.983  |
|        |      | 7.5               | 0.973    | 0.818          | 0.027        | 0.986    | 0.804  |
|        |      | 15                | 0.958    | 0.889          | 0.042        | 0.982    | 0.829  |
|        |      | 25                | 0.973    | 0.933          | 0.027        | 0.978    | 0.928  |

**Tabela complementar S4.** Métricas de desempenho individuais obtidas na etapa de validação do GroundEye S800, para as amostras de sementes com presença de defeito tipo popped kernel nas proporções de 0%, 2,5%, 7,5%, 15% e 25%. (Continua)

| Método         | Perc | Preceça de dano % | Acurácia | Especificidade | Taxa de erro | Precisão | MCC   |
|----------------|------|-------------------|----------|----------------|--------------|----------|-------|
| Kruskal-Wallis | 0.05 | 0                 | 0.955    | 0.667          | 0.045        | 0.998    | 0.208 |
|                |      | 2.5               | 0.948    | 0.824          | 0.052        | 0.995    | 0.504 |
|                |      | 7.5               | 0.937    | 0.886          | 0.063        | 0.991    | 0.663 |
|                |      | 15                | 0.915    | 0.827          | 0.085        | 0.972    | 0.682 |
|                |      | 25                | 0.945    | 0.947          | 0.055        | 0.982    | 0.861 |
|                | 0.1  | 0                 | 0.975    | 0.667          | 0.025        | 0.998    | 0.282 |
|                |      | 2.5               | 0.972    | 0.824          | 0.028        | 0.995    | 0.629 |
|                |      | 7.5               | 0.96     | 0.864          | 0.04         | 0.989    | 0.745 |
|                |      | 15                | 0.943    | 0.901          | 0.057        | 0.984    | 0.784 |
|                |      | 25                | 0.955    | 0.907          | 0.045        | 0.969    | 0.88  |
|                | 0.2  | 0                 | 0.98     | 0.667          | 0.02         | 0.998    | 0.314 |
|                |      | 2.5               | 0.978    | 0.824          | 0.022        | 0.995    | 0.683 |
|                |      | 7.5               | 0.967    | 0.795          | 0.033        | 0.984    | 0.76  |
|                |      | 15                | 0.953    | 0.889          | 0.047        | 0.982    | 0.812 |
|                |      | 25                | 0.963    | 0.92           | 0.037        | 0.973    | 0.902 |
|                | 0.3  | 0                 | 0.993    | 1              | 0.007        | 1        | 0.652 |
|                |      | 2.5               | 0.987    | 0.824          | 0.013        | 0.995    | 0.772 |
|                |      | 7.5               | 0.973    | 0.795          | 0.027        | 0.984    | 0.8   |
|                |      | 15                | 0.958    | 0.889          | 0.042        | 0.982    | 0.892 |
|                |      | 25                | 0.973    | 0.927          | 0.027        | 0.976    | 0.928 |
|                | 0.5  | 0                 | 0.988    | 0.667          | 0.012        | 0.998    | 0.404 |
|                |      | 2.5               | 0.98     | 0.824          | 0.02         | 0.995    | 0.698 |
|                |      | 7.5               | 0.977    | 0.818          | 0.023        | 0.986    | 0.825 |
|                |      | 15                | 0.957    | 0.889          | 0.043        | 0.982    | 0.823 |
|                |      | 25                | 0.975    | 0.933          | 0.025        | 0.978    | 0.933 |
|                | 0.75 | 0                 | 0.993    | 0.667          | 0.007        | 0.998    | 0.513 |
|                |      | 2.5               | 0.982    | 0.824          | 0.018        | 0.995    | 0.715 |
|                |      | 7.5               | 0.978    | 0.818          | 0.022        | 0.986    | 0.836 |
|                |      | 15                | 0.965    | 0.889          | 0.035        | 0.983    | 0.853 |
|                |      | 25                | 0.975    | 0.933          | 0.025        | 0.974    | 0.933 |
|                | 1    | 0                 | 0.985    | 0.667          | 0.015        | 0.998    | 0.36  |
|                |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.983 |
|                |      | 7.5               | 0.973    | 0.818          | 0.027        | 0.986    | 0.804 |
|                |      | 15                | 0.958    | 0.889          | 0.042        | 0.982    | 0.829 |
|                |      | 25                | 0.973    | 0.933          | 0.027        | 0.978    | 0.928 |

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| Método    | Perc | Presença de dano % | Acurácia | Especificidade | Taxa de erro | Precisão | MCC   |
|-----------|------|--------------------|----------|----------------|--------------|----------|-------|
| Variância | 0.05 | 0                  | 0.85     | 0.333          | 0.15         | 0.996    | 0.037 |
|           |      | 2.5                | 0.853    | 0.588          | 0.147        | 0.986    | 0.208 |
|           |      | 7.5                | 0.83     | 0.727          | 0.17         | 0.975    | 0.366 |
|           |      | 15                 | 0.8      | 0.679          | 0.2          | 0.945    | 0.394 |
|           |      | 25                 | 0.823    | 0.707          | 0.177        | 0.898    | 0.549 |
|           | 0.1  | 0                  | 0.933    | 0.667          | 0.067        | 0.998    | 0.168 |
|           |      | 2.5                | 0.92     | 0.824          | 0.08         | 0.994    | 0.416 |
|           |      | 7.5                | 0.913    | 0.705          | 0.087        | 0.975    | 0.515 |
|           |      | 15                 | 0.88     | 0.802          | 0.12         | 0.967    | 0.592 |
|           |      | 25                 | 0.913    | 0.853          | 0.087        | 0.95     | 0.773 |
|           | 0.2  | 0                  | 0.98     | 0.667          | 0.02         | 0.998    | 0.314 |
|           |      | 2.5                | 0.982    | 0.824          | 0.018        | 0.995    | 0.715 |
|           |      | 7.5                | 0.963    | 0.818          | 0.037        | 0.985    | 0.748 |
|           |      | 15                 | 0.937    | 0.84           | 0.063        | 0.974    | 0.747 |
|           |      | 25                 | 0.948    | 0.873          | 0.052        | 0.958    | 0.86  |
|           | 0.3  | 0                  | 0.99     | 0.667          | 0.01         | 0.988    | 0.432 |
|           |      | 2.5                | 0.985    | 0.824          | 0.015        | 0.995    | 0.752 |
|           |      | 7.5                | 0.977    | 0.841          | 0.023        | 0.987    | 0.828 |
|           |      | 15                 | 0.958    | 0.852          | 0.042        | 0.977    | 0.823 |
|           |      | 25                 | 0.965    | 0.9            | 0.035        | 0.967    | 0.906 |
|           | 0.5  | 0                  | 0.985    | 0.33           | 0.015        | 0.997    | 0.198 |
|           |      | 2.5                | 0.978    | 0.824          | 0.022        | 0.995    | 0.683 |
|           |      | 7.5                | 0.97     | 0.795          | 0.03         | 0.984    | 0.779 |
|           |      | 15                 | 0.963    | 0.877          | 0.037        | 0.981    | 0.845 |
|           |      | 25                 | 0.973    | 0.933          | 0.027        | 0.978    | 0.928 |
|           | 0.75 | 0                  | 0.992    | 0.667          | 0.008        | 0.998    | 0.468 |
|           |      | 2.5                | 0.978    | 0.824          | 0.022        | 0.995    | 0.683 |
|           |      | 7.5                | 0.975    | 0.818          | 0.025        | 0.986    | 0.814 |
|           |      | 15                 | 0.958    | 0.889          | 0.042        | 0.982    | 0.829 |
|           |      | 25                 | 0.915    | 0.927          | 0.032        | 0.976    | 0.915 |
|           | 1    | 0                  | 0.985    | 0.667          | 0.015        | 0.998    | 0.36  |
|           |      | 2.5                | 0.983    | 0.824          | 0.017        | 0.995    | 0.983 |
|           |      | 7.5                | 0.973    | 0.818          | 0.027        | 0.986    | 0.804 |
|           |      | 15                 | 0.958    | 0.889          | 0.042        | 0.982    | 0.829 |
|           |      | 25                 | 0.973    | 0.933          | 0.027        | 0.978    | 0.928 |

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| Método              | Perc | Preceça de dano % | Acurácia | Especificidade | Taxa de erro | Precisão | MCC   |
|---------------------|------|-------------------|----------|----------------|--------------|----------|-------|
| Ganho de Informação | 0.05 | 0                 | 0.96     | 0.667          | 0.04         | 0.998    | 0.222 |
|                     |      | 2.5               | 0.95     | 0.824          | 0.05         | 0.995    | 0.511 |
|                     |      | 7.5               | 0.94     | 0.884          | 0.06         | 0.991    | 0.67  |
|                     |      | 15                | 0.92     | 0.815          | 0.08         | 0.97     | 0.692 |
|                     |      | 25                | 0.945    | 0.953          | 0.055        | 0.984    | 0.862 |
|                     | 0.1  | 0                 | 0.977    | 0.667          | 0.023        | 0.998    | 0.291 |
|                     |      | 2.5               | 0.968    | 0.824          | 0.032        | 0.995    | 0.606 |
|                     |      | 7.5               | 0.973    | 0.907          | 0.027        | 0.993    | 0.973 |
|                     |      | 15                | 0.952    | 0.901          | 0.048        | 0.84     | 0.809 |
|                     |      | 25                | 0.953    | 0.927          | 0.047        | 0.975    | 0.877 |
|                     | 0.2  | 0                 | 0.987    | 0.667          | 0.013        | 0.998    | 0.38  |
|                     |      | 2.5               | 0.978    | 0.824          | 0.022        | 0.995    | 0.683 |
|                     |      | 7.5               | 0.972    | 0.841          | 0.028        | 0.987    | 0.798 |
|                     |      | 15                | 0.952    | 0.864          | 0.048        | 0.979    | 0.801 |
|                     |      | 25                | 0.97     | 0.933          | 0.03         | 0.978    | 0.92  |
|                     | 0.3  | 0                 | 0.987    | 0.667          | 0.013        | 0.998    | 0.38  |
|                     |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.733 |
|                     |      | 7.5               | 0.973    | 0.795          | 0.027        | 0.984    | 0.8   |
|                     |      | 15                | 0.957    | 0.901          | 0.043        | 0.984    | 0.826 |
|                     |      | 25                | 0.972    | 0.933          | 0.028        | 0.978    | 0.924 |
|                     | 0.5  | 0                 | 0.988    | 0.667          | 0.012        | 0.998    | 0.404 |
|                     |      | 2.5               | 0.978    | 0.824          | 0.022        | 0.995    | 0.683 |
|                     |      | 7.5               | 0.973    | 0.841          | 0.027        | 0.987    | 0.808 |
|                     |      | 15                | 0.953    | 0.877          | 0.047        | 0.98     | 0.809 |
|                     |      | 25                | 0.97     | 0.92           | 0.03         | 0.974    | 0.919 |
|                     | 0.75 | 0                 | 0.99     | 0.667          | 0.01         | 0.998    | 0.432 |
|                     |      | 2.5               | 0.988    | 0.824          | 0.012        | 0.995    | 0.794 |
|                     |      | 7.5               | 0.982    | 0.83           | 0.018        | 0.986    | 0.868 |
|                     |      | 15                | 0.957    | 0.889          | 0.043        | 0.982    | 0.823 |
|                     |      | 25                | 0.975    | 0.933          | 0.025        | 0.978    | 0.933 |
|                     | 1    | 0                 | 0.983    | 0.5            | 0.017        | 0.997    | 0.309 |
|                     |      | 2.5               | 0.983    | 0.824          | 0.017        | 0.995    | 0.983 |
|                     |      | 7.5               | 0.973    | 0.818          | 0.027        | 0.986    | 0.804 |
|                     |      | 15                | 0.958    | 0.889          | 0.042        | 0.982    | 0.829 |
|                     |      | 25                | 0.973    | 0.933          | 0.027        | 0.978    | 0.928 |